

HarvestTrust

Every delivery recorded. Every payment explained.

A Transparent Produce Collection & Payment Register for Farmer Producer Groups

This digital system replaces traditional paper slips with an immutable SQLite-based collection register, verifies pricing on the server, tracks outstanding dues in a bank-ledger format, and flags potential dispute cases using a Random Forest machine learning classifier.

Candidate Details:

Student Name: JANARTHANAN V (Register Number: 411723205021)

Institution/Department: PSVPEC - Department of Information Technology - Year IV

Assessment Level: Easy (SIH 2026 Practical Assessment)

1. The Trust Problem in Farmer Groups

The inefficiencies of paper-slip collections

- **Slow Transactions:** Recording delivery info on paper slips causes long queues during harvest times.
- **Calculation Inaccuracies:** Manual math on multiple slips leads to operator errors in weights and final totals.
- **Disputes & Lack of Transparency:** Farmers cannot easily verify their delivered history or clarify unpaid dues, leading to loss of trust.
- **Delayed Payments:** Secretaries spend days aggregating paper bundles to calculate pay registers, delaying farmer payments.

2. Primary Stakeholders Affected

Who benefits from the digital transformation

- **Collection Operator:** Needs a quick, clean interface to weigh produce, select farmers, and save records instantly without doing manual math.
- **Farmer / Member:** Deserves a transparent, print-ready transaction statement showing opening balance, deliveries, payments, and closing balances.
- **Group Secretary:** Requires a central dashboard to review attention-flagged anomalies, record payments, and audit material log changes.

3. The Solution: HarvestTrust

A digital ledger and decision engine for agriculture

- **Digital Capture:** Replaces paper with a simple React-Vite web app connected to a central Express/Prisma server.
- **Authoritative Pricing:** Final amount computed securely on the server to prevent operator tampering or error.
- **Account Statement:** Full bank-style ledger tracks outstanding balances for each farmer dynamically.
- **Risk Engine:** Machine learning predicts potential disputes or anomalous inputs at collection time.

4. End-to-End Collection Flow

Recording a new produce collection with server validation

The Operator enters the farmer code, selects the produce type, and inputs the scale weight.

Authoritative server-side calculations: - Auto-fills base rates according to effective rate history. - Gross & Net amounts are calculated on the backend using decimal-safe rounding. - Generates a unique, readable slip receipt number (e.g. *HT-20260724-0006*).

The screenshot displays the 'New Collection' interface of the HarvestTrust application. On the left is a dark green sidebar with navigation options: Dashboard, New Collection (highlighted), Register, Members, Payments, Attention Queue, and Reports. The user is logged in as JANARTHANAN V (Admin). The main content area shows a 'New Collection' header with a 'New Collection' button and 'ADMIN MODE' indicator. A central white card displays a green checkmark icon and the message 'COLLECTION SAVED AUTHORITATIVELY Slip Generated'. Below this, the receipt number 'HT-20260724-0006' is shown. A table lists the collection details: Farmer (Krishnan M (MEM001)), Produce (Paddy), Quantity (40 kg), Rate (₹22.50 / kg), and Authoritative Total (₹900.00). At the bottom of the card are buttons for 'Record Another Delivery', 'Open Member Statement', and a link to 'Go to Collection Register'.

COLLECTION SAVED AUTHORITATIVELY	
Slip Generated	
Receipt Number: HT-20260724-0006	
Farmer:	Krishnan M (MEM001)
Produce:	Paddy
Quantity:	40 kg
Rate:	₹22.50 / kg
Authoritative Total:	₹900.00

5. Server-Side Calculation Verification

Validating pricing math and SQLite constraint transactions

Authoritative Formula:

$\text{grossAmount} = \text{round}(\text{quantity} \times \text{ratePerUnit}, 2)$

Verification Hand Math:

Quantity: **125.50 kg** | Rate: **32.40 / kg**

Calculated: $125.50 \times 32.40 = \mathbf{4,066.20}$

Transaction Rollback & SQLite Security:

- Reject zero/negative quantity or rate. Reject duplicate receipts.
- Rollback database if any related operation fails during multi-row insert.

6. Prioritizing Attention Cases first

Machine learning decision assistance and threshold rules

A Random Forest model classifier runs live before submission on pre-outcome inputs: - Flags high moisture (e.g. >18%), odd hours, or quality/rate conflicts. - Confidence threshold rule set at **65%**. - Low-confidence predictions produce no class, indicating *Review Normally*. - High-risk cases are ordered first in the register and sent to the Secretary's review queue.

The screenshot displays the HarvestTrust 'Register' interface. On the left is a dark green sidebar menu with options: Dashboard, New Collection, Register (highlighted), Members, Payments, Attention Queue, and Reports. At the bottom of the sidebar, the user is identified as JANARTHANAN V (Admin) with a Logout button. The main content area has a top bar with 'New Collection' and 'ADMIN MODE' buttons. Below this is a search bar containing 'HT-20260724-0006' and a 'Clear All Filters' button. A row of filters includes: PRODUCE (All Produce), POINT (All Points), PAYMENT (All Statuses), ATTENTION (All Risk Cases), FROM DATE (mm/dd/yyyy), and TO DATE (mm/dd/yyyy). The main table lists collection records with columns: RECEIPT, DATE, FARMER NAME, PRODUCE, WEIGHT, RATE, AMOUNT, PAYMENT, RISK STATUS, and ACTION. The first record shown is for receipt HT-20260724-0006, dated 24 Jul 2026, from farmer Krishnan M (MEM0011, Melpuram), for 40 kg of Paddy at a rate of ₹22.50, totaling ₹900.00, with a status of UNPAID and a risk status of NONE. The footer of the table indicates 'Showing Page 1 of 11 (110 records total)'.

RECEIPT	DATE	FARMER NAME	PRODUCE	WEIGHT	RATE	AMOUNT	PAYMENT	RISK STATUS	ACTION
HT-20260724-0006	24 Jul 2026	Krishnan M MEM0011 Melpuram	Paddy	40 kg	₹22.50	₹900.00	UNPAID	NONE	

7. HarvestTrust Outcomes & Next Improvements

Current implementation status and future enhancements

Working Today:

- ✓ End-to-end receipt collection, validation, and storage.
- ✓ Chronological member statements calculating running balances.
- ✓ Random Forest risk classifier (Accuracy 79%).
- ✓ CSV summary exports, print stylesheet layouts, and automated test reports.

Next Improvements:

1. **Farmer SMS/WhatsApp Integrations:** Send instant receipts/ledger statement updates.
2. **Offline-First Support:** Allow operators to record weights offline during network failures.